Experimental Design for Dynamic Treatment Regimes

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PubH 7485/8485

Sequential Multiple Assignment Randomized Trial

- SMART Designs are type of trial design that scientists can use to optimize a DTR
- Can be used to answer any of the following questions

Sequential Multiple Assignment Randomized Trial

- Multi-stage trial design -> each stage corresponds to a critical decision
- Participant can be randomly assigned to one of several intervention options at each stage
- Set of available interventions that a single participant may be randomized to at each stage may depend on prior treatment received and/or participant characteristics
- In a SMART, some or all participants are randomized at least twice; i.e., at two or more stages

Case Study #1: PLUTO

- Several behavioral, pharmacological, and combination treatment strategies for smoking cessation
- Treatments yield low rates of long-term abstinence (5-20%), and there is little evidence to help development of an AIS
- Pressing need: USPSTF recommends annual lung cancer screening for those age 55 to 80 years who have a 30 or more pack year history of smoking and who are current smokers or have quit within the last 15 years

Case Study #1: PLUTO

- Usual care: 8 weeks of telephone counseling + NRT
- Tobacco longitudinal care: continue monthly calls for one year -> recycle relapsers and reducers to make additional quit attempts
- TLC algorithm for treatment non-responders

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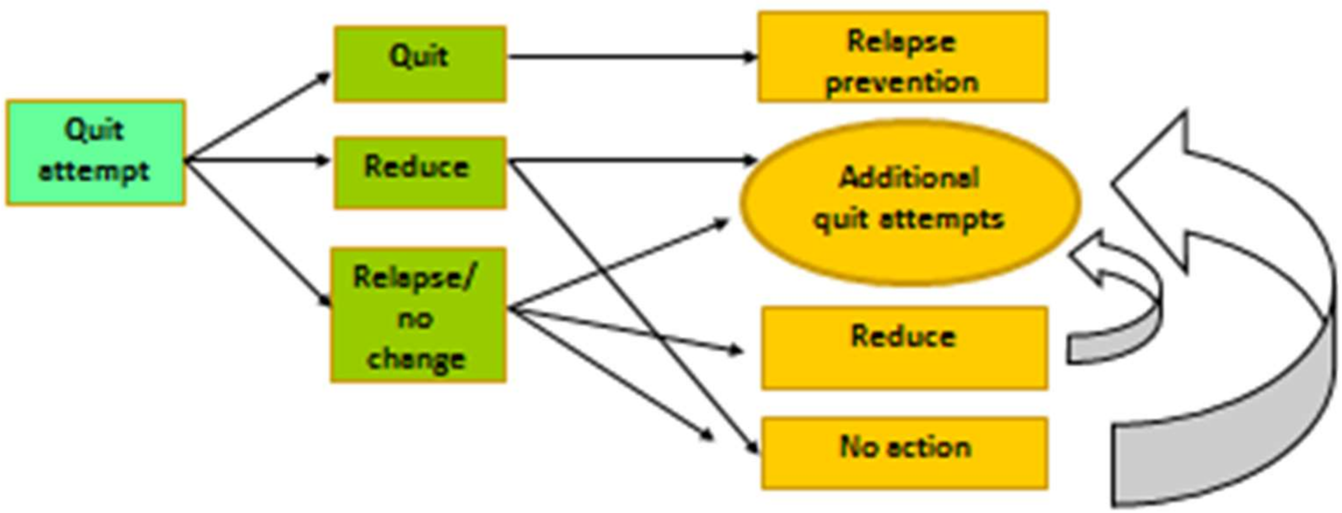
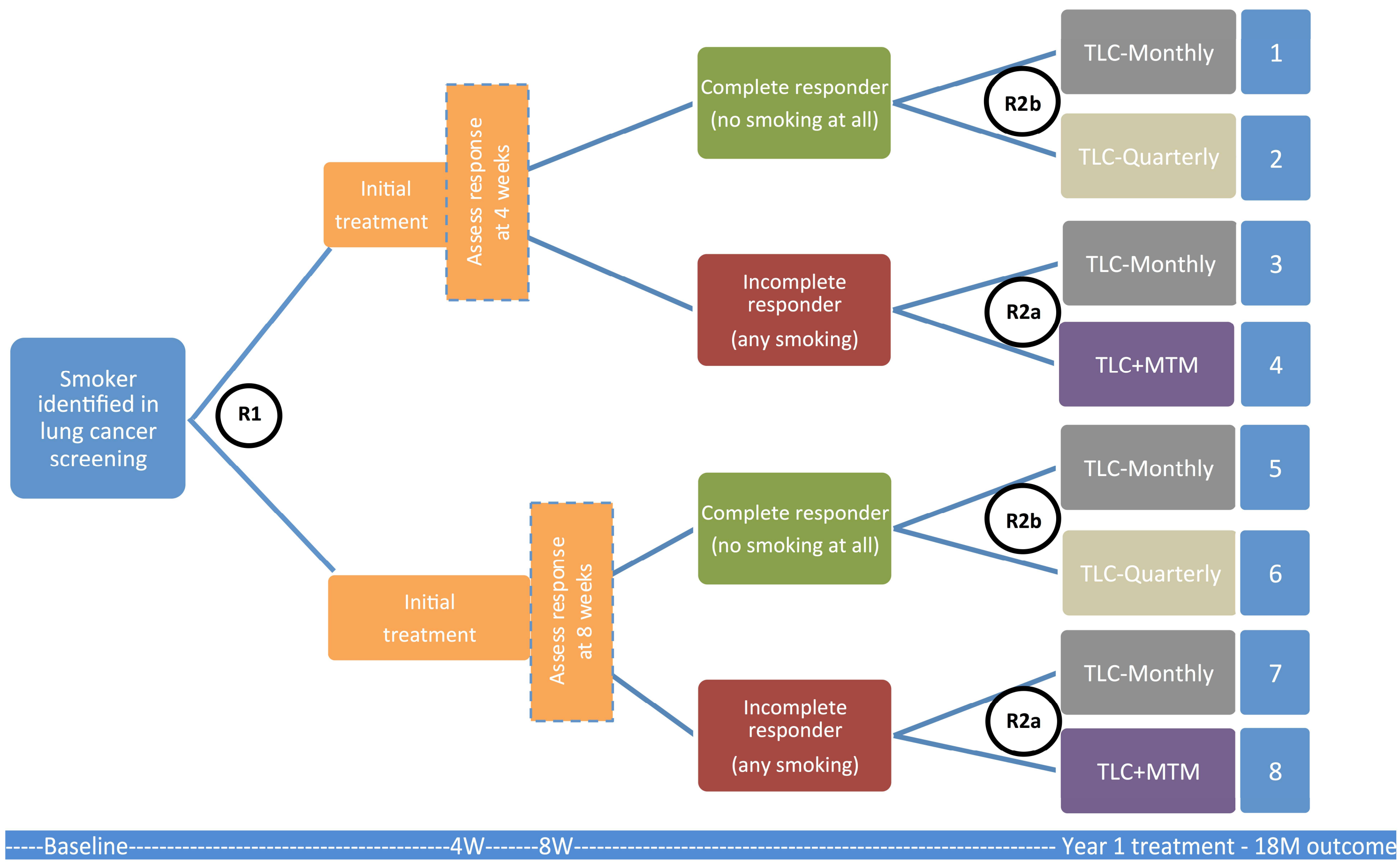


Case Study #1: PLUTO

- TLC is an intensive intervention compared to state quit lines
- Other interventions which are evidence-based which could help those who are struggling
- Key questions:

1) For those "doing well" can we reduce the frequency of phone counseling
2) For those "struggling" will offering pharmacotherapy improve outcomes?

Case Study #1: PLUTO



Embedded Tailoring Variable/DTR

- Response or nonresponse status at the end of initial-phase TLC is used to tailor the interventions that a subject may be randomized to in the second-phase
- Because this tailoring variable is part of the study design ->??? "embedded tailoring variable"
- DTR which use this tailoring variable and intervention components in the study are known as "embedded adaptive intervention"

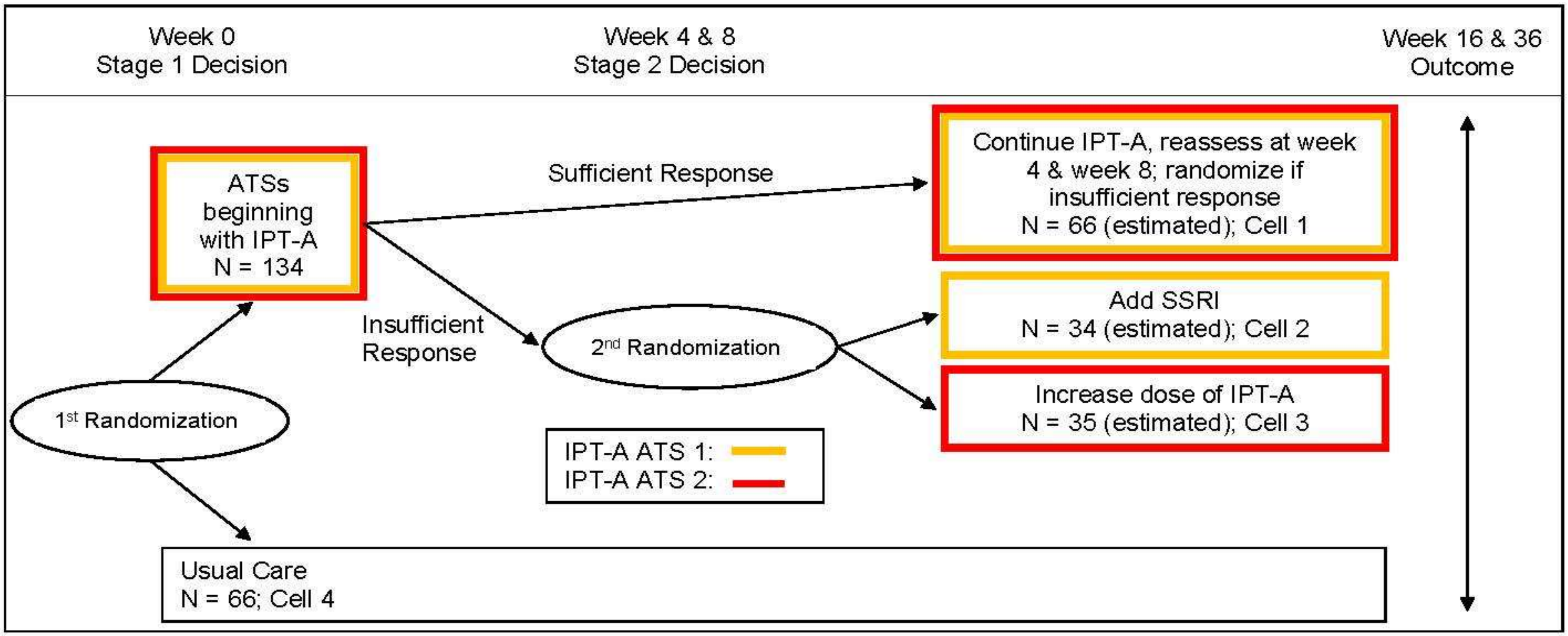
Embedded AIS in PLUTO

	Phase 1 Intervention	Phase 2 Intervention for Responders	Phase 2 Intervention for Nonresponders	Cells in Figure 1 Consistent with this DTR
1	4 weeks of Initial Phase of TLC	TLC-Monthly	TLC-Monthly	1 & 3
2	4 weeks of Initial Phase of TLC	TLC-Monthly	TLC+MTM	1 & 4
3	4 weeks of Initial Phase of TLC	TLC-Quarterly	TLC-Monthly	2 & 3
4	4 weeks of Initial Phase of TLC	TLC-Quarterly	TLC+MTM	2 & 4
5	8 weeks of Initial Phase of TLC	TLC-Monthly	TLC-Monthly	5 & 7
6	8 weeks of Initial Phase of TLC	TLC-Monthly	TLC+MTM	5 & 8
7	8 weeks of Initial Phase of TLC	TLC-Quarterly	TLC-Monthly	6 & 7
8	8 weeks of Initial Phase of TLC	TLC-Quarterly	TLC+MTM	6 & 8

Case Study #2: Algorithms for Adolescent Depression

- 20% of adolescents experiencing a depressive episode at some point
- Several evidence-based treatments including psychotherapy (interpersonal psychotherapy, cognitive behavioral therapy), antidepressant medication (SSRIs), and their combination.
- Approximately 30-50% of adolescents who receive these treatments do not respond
- Practice parameters recommend systematic and routine assessment and monitoring
- Currently no guidelines to direct therapists regarding how to use those symptom assessments to guide subsequent treatment decisions

Case Study #2: Algorithms for Adolescent Depression



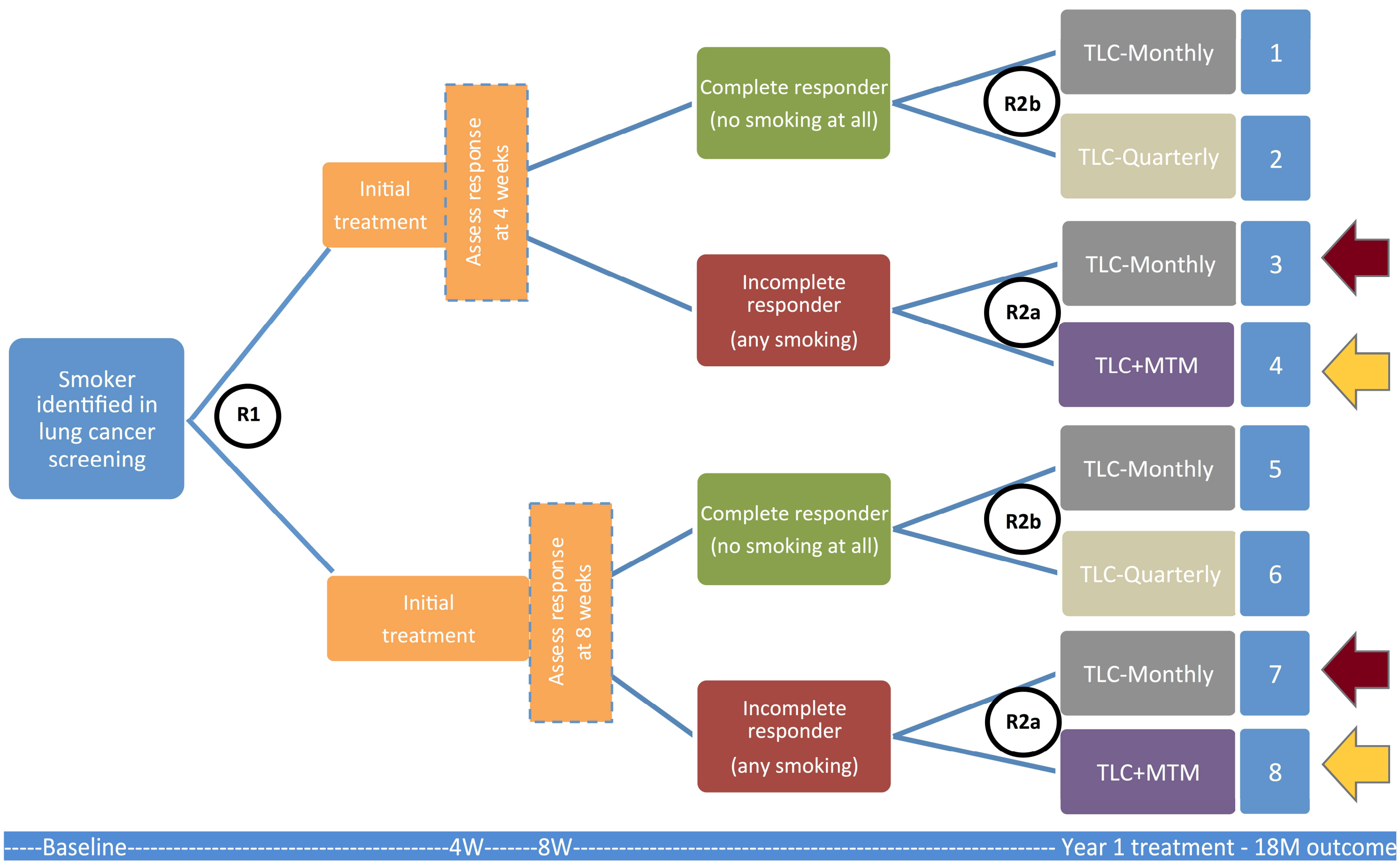
Specific Aims of PLUTO

- Primary: Test whether incomplete responders to initial phase of TLC treatment benefit from the addition of MTM
- Secondary: Among complete responders compare the effect of more intensive (at least monthly) follow-up to less intensive (quarterly) follow-up
- Exploratory: Moderators of second-phase treatment (assessment time & # of days smoking in last week)
- Exploratory: Compare different embedded adaptive interventions
- Note: primary and secondary aim mirror the key questions outlined earlier

Stat Analysis and Power for Primary Aim

- As with factorial designs, common test the main effects of intervention components
- To test primary aim of PLUTO: pool data from all incomplete responders randomized to TLC (cells 3 & 7) or TLC+MTM (cells 4 & 8 and compare the outcomes using methods appropriate for the comparison of two groups
- Standard formulas can be used to estimate the sample size of nonresponders needed
- Overall study sample size: Dividing the sample size of nonresponders by the proportion of subjects one anticipates to be nonresponders

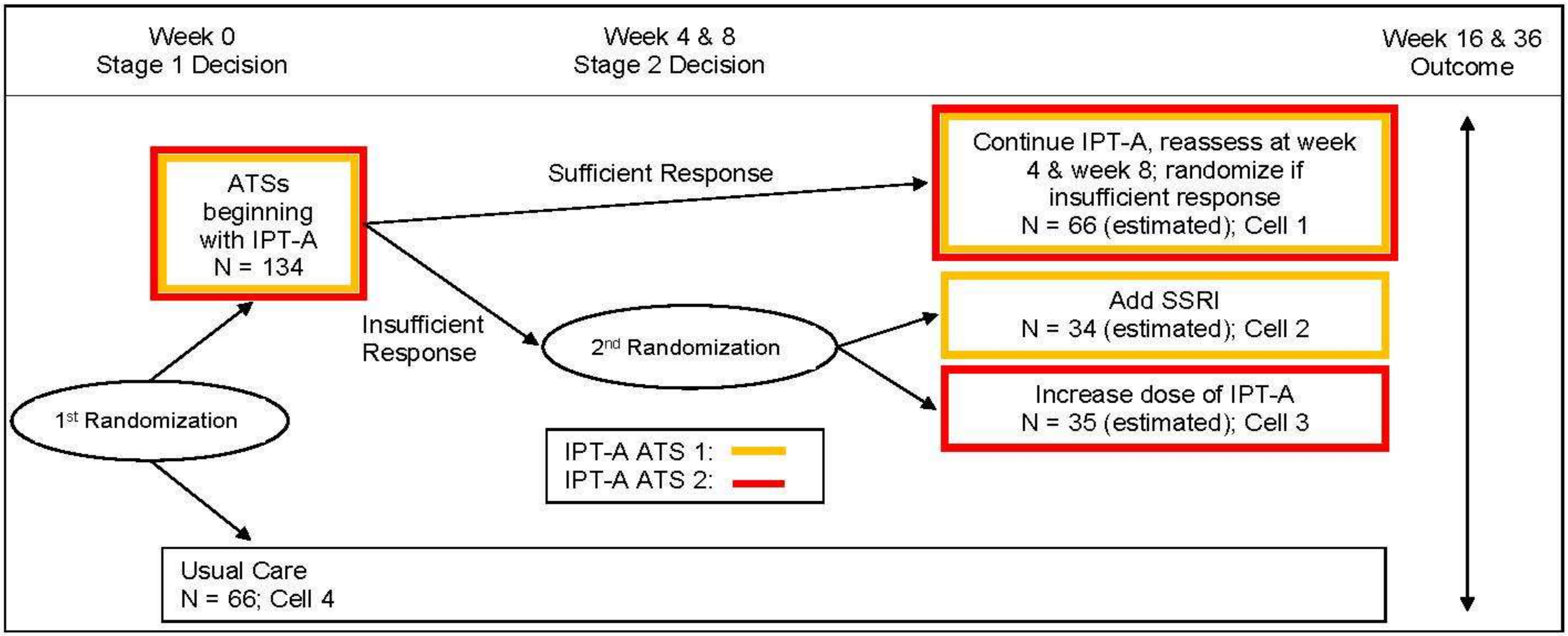
Case Study #1: PLUTO



Stat Analysis to Compare DTRs

- Want to estimate the average response to the DTR "begin with IPT-A and nonresponders receive IPT-A + SSRI"
- Data from participants consistent with that embedded DTR (i.e., participants who follow the treatment paths ending in cells 1 and 2) would be used
- Subjects are compliant with more than one embedded DTR. Therefore, estimators of the anticipated outcome of different embedded DTRs may be correlated ??? robust standard errors
- Responders who are consistent with an embedded AIS are over- or underrepresented relative to what we would expect if everyone in the population were to follow this regimen. Over- or underrepresentation in a sample can be corrected using a weighted analysis

Case Study #2: Algorithms for Adolescent Depression



Design Considerations

- Old adage: "Protocolize what you know, randomize what you don't know"
- Lots to consider as part of SMART
- Initial phase treatment type, intensity, and duration
- Tailoring variable, cutpoint, and assessment time
- Second phase treatment type, intensity, and duration for both "responders" and "nonresponders"
- Cannot randomize them all and usually infeasible to randomize to more than 2-3 levels

Control Conditions Within a SMART

- Many SMARTs do not include a “control” or “unpersonalized” condition -> goal is to test efficacy of several AIS one of which could be carried forward
- SMARTs can be designed to include a suitable control intervention
  1) As part of the embedded AIS
  2) As part of the initial randomization
- Effect sizes will be smaller comparing two different high quality AISs than comparing AIS to usual care

Control AIS in PLUTO

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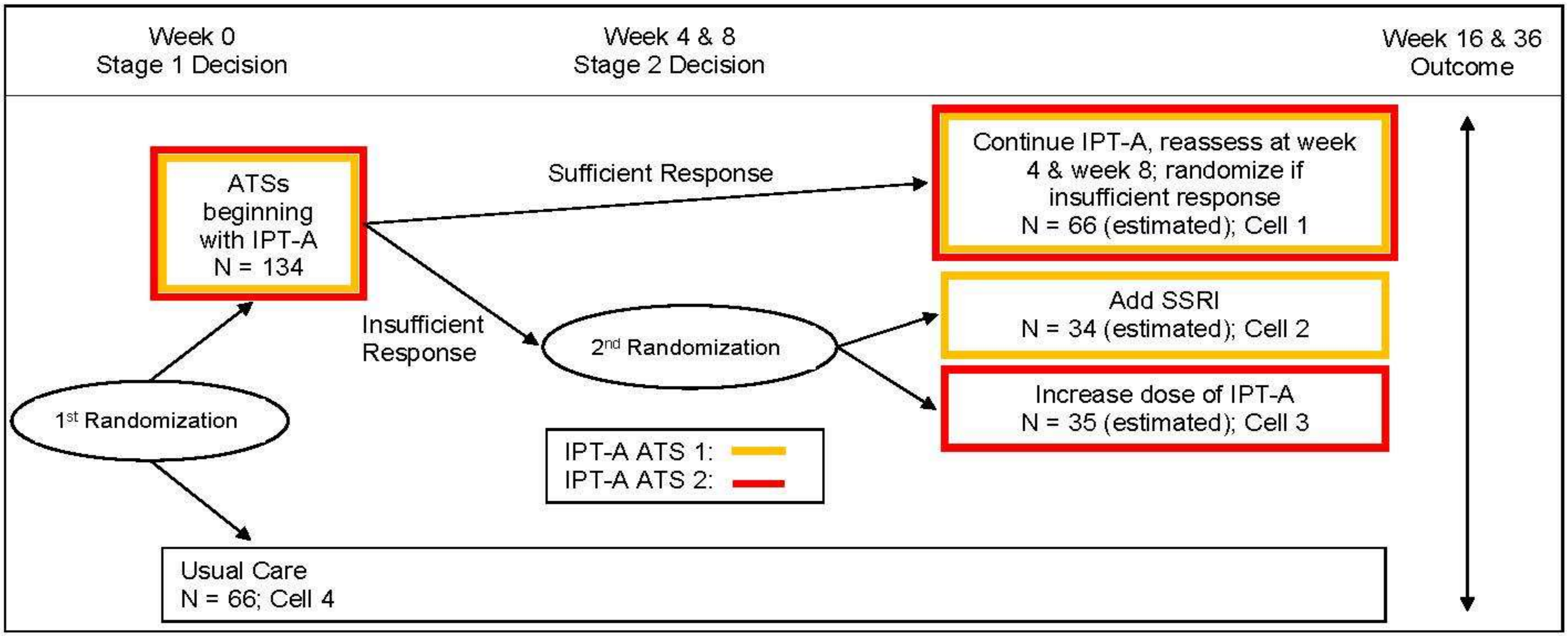
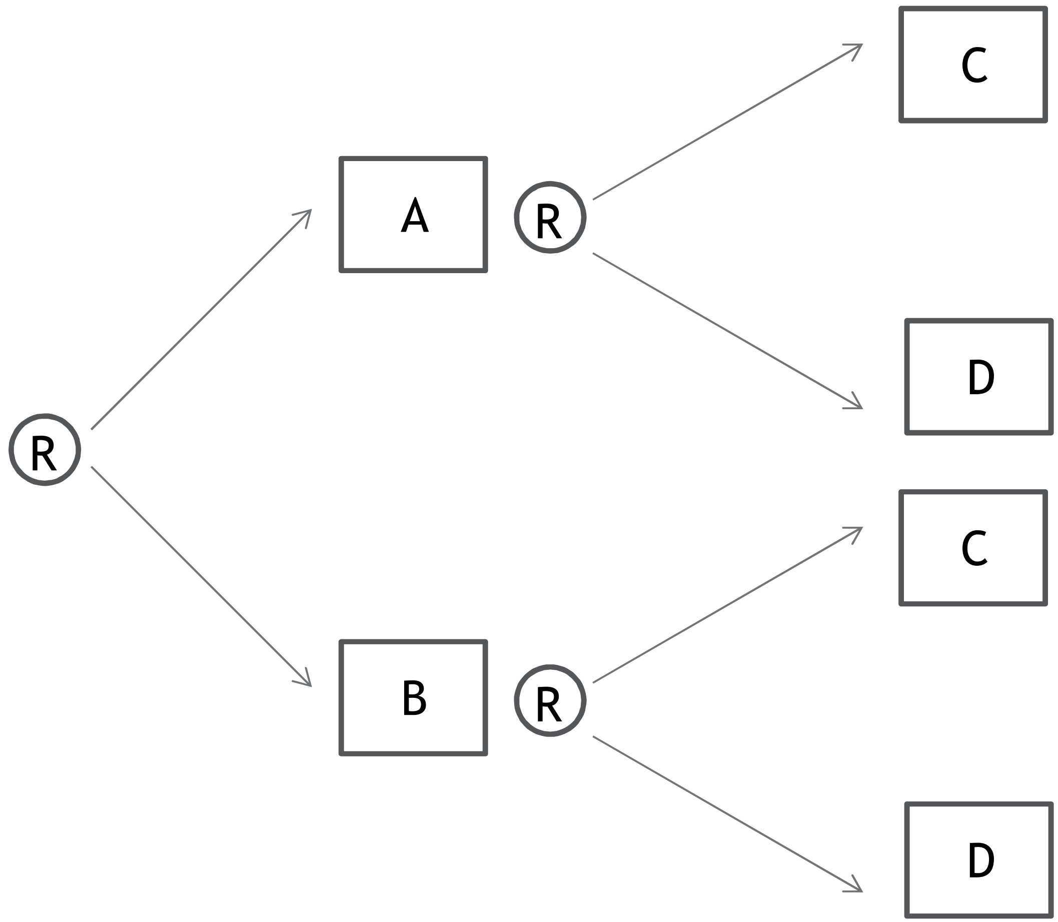
Case Study #2: Algorithms for Adolescent Depression



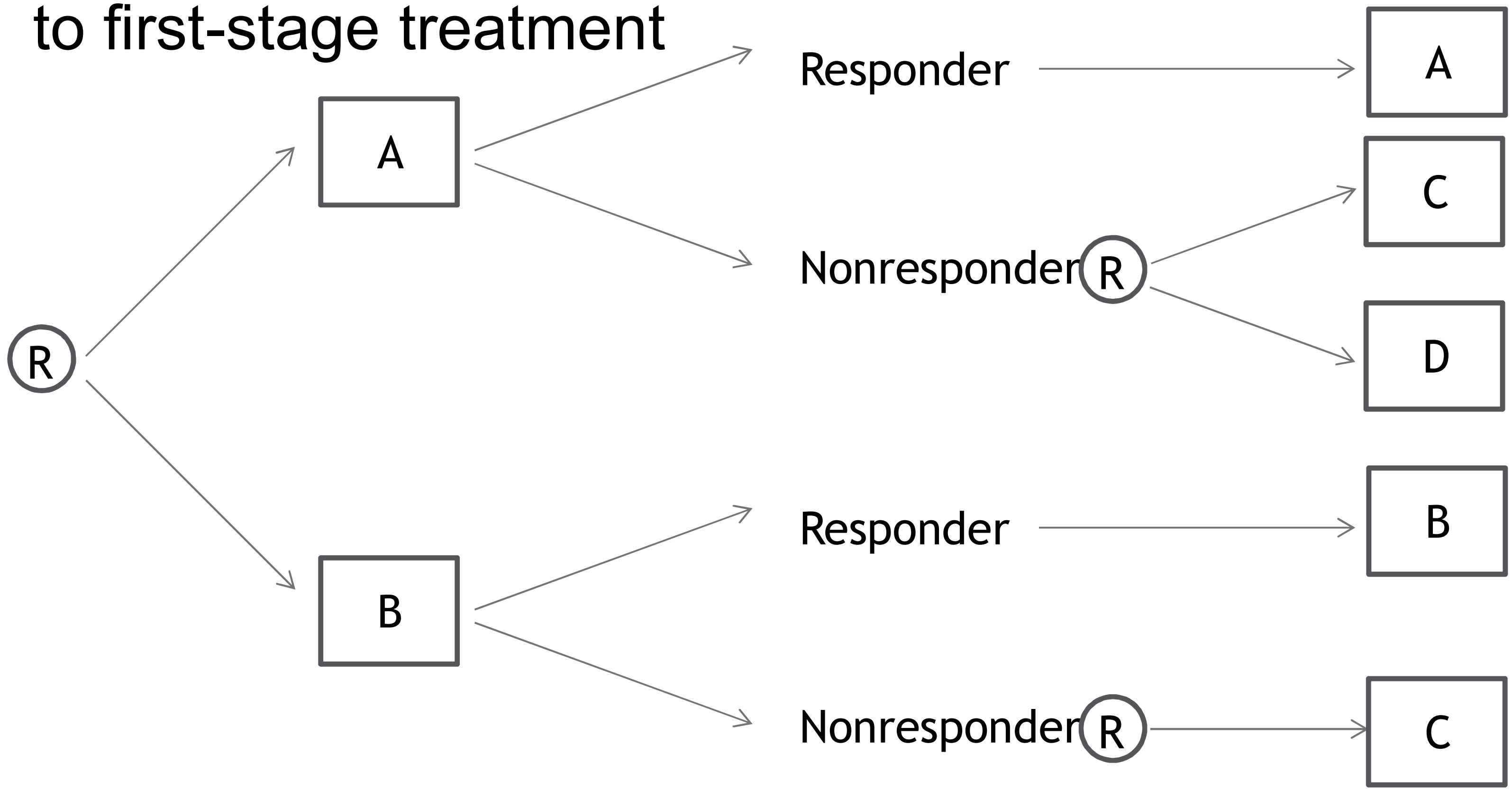
Other Common SMART Designs

- SMART with no embedded tailoring variable



Other Common SMART Designs

- SMART in which re-randomization depends on a combination of first-stage treatment and nonresponse to first-stage treatment



Other Experimental Designs

- Factorial Design
- Multi-arm RCT
- Standard RCT or series of two-arm RCTs comparing different adaptive intervention strategies
- Single-stage-at-a-time experimental approach (e.g., nonresponder trial, discontinuation trial, etc.)

Other Experimental Designs: Drawbacks

- Factorial Design - Does not allow embedded tailoring variables
- Multi-arm RCT - Requires larger sample size
- Standard RCT or series of two-arm RCTs comparing different adaptive intervention strategies - Requires MUCH larger sample size
- Single-stage-at-a-time experimental approach (e.g., nonresponder trial, discontinuation trial, etc.) - 1) interventions may have delayed effects or interact with later-stage treatments 2) the types of nonresponding subjects who agree to participant in a trial testing two second-stage treatments may differ from the population of nonresponders

Common Misconceptions of DTR

- Tailoring variables cannot differ based on previous intervention
- A DTR must recommend a single intervention component at each decision point
- DTRs seek to replace clinical judgement
- DTRs are only relevant in treatment settings
- DTRs are non-standard because they involve randomization

Common Misconceptions of SMART

- SMARTs require prohibitively large sample sizes
- All SMARTs require Multiple-Comparisons Adjustments
- All research on adaptive interventions requires a SMART
- All SMARTs must include an embedded tailoring variable
- All aspects of an embedded adaptive intervention must be randomized
- SMARTs are a form of adaptive research design
- SMARTs never include a control group
- SMARTs require multiple consents
- SMARTs are susceptible to high levels of study drop-out